

TAGSNav: Target-Aware Online 3D Gaussian Splatting and Active Exploration for Instance ImageGoal Navigation

Abstract—Instance ImageGoal Navigation (IIN) is a fundamental embodied navigation task requiring an agent to locate and navigate to a specific object instance in an unknown environment given a reference image. Its core challenge is recognizing the same target instance across viewpoints while distinguishing it from visually similar instances of the same category under sparse observations and computational constraints. Most existing methods rely on 2D semantic maps or scene graphs, with target appearance representations limited to observed viewpoints, restricting instance matching across arbitrary viewpoints. 3D Gaussian Splatting (3DGS) can alleviate this limitation through novel-view rendering for viewpoint-aligned matching. However, optimization-based online 3DGS still struggles to preserve reconstruction details for instance-level discrimination under sparse observations and limited optimization budgets. To this end, we propose TAGSNav, an online IIN framework unifying target-aware online Gaussian mapping and instance verification in a shared scene representation. Its closed mapping–exploration loop couples incremental Semantic Keyframe-Guided Gaussian Splatting (SKGS) mapping for target-region optimization with Gaussian-uncertainty-driven Semantically Weighted Mutual Information (SWMI)-guided exploration for informative viewpoint selection, improving online local reconstruction under limited observations and computation. Building on these maps, coarse-to-fine localization conditionally invokes viewpoint alignment. If the target remains unlocalized, compensatory re-exploration revisits candidates; together, they reduce false and missed matches across arbitrary viewpoints. On HM3D-SEM, it outperforms state-of-the-art methods in SR and SPL; physical-robot deployment without re-training further demonstrates Sim2Real generalization. Project page: <https://tagsnav754.github.io/TAGSNav/>.

I. INTRODUCTION

Instance ImageGoal Navigation (IIN) requires an embodied agent to explore an unknown environment and navigate to a specific object instance depicted in a reference image, which is an important task in embodied visual navigation [1], [2]. Image-based target specification provides an intuitive interface for user-oriented object search by mobile robots in complex indoor environments [3]. However, online IIN remains challenging when the reference image is captured from an extreme viewpoint, as the robot must verify the target instance from sparse observations while rejecting visually similar distractors [4].

Existing IIN methods fall into end-to-end and modular approaches. End-to-end methods [4] map observations directly to actions without explicit environmental representations, limiting interpretability and controllability. Modular methods [5], [6] maintain explicit representations for mapping and navigation, but their coarse 2D semantic maps or scene graphs cannot adequately preserve the fine-grained appearance and 3D geometry required for instance matching.

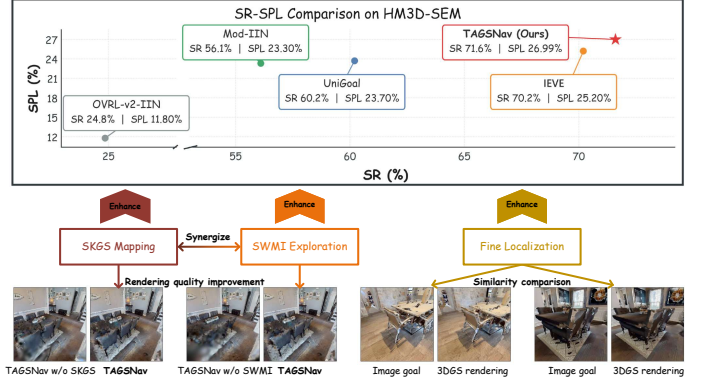


Fig. 1: Overview and performance of TAGSNav on HM3D-SEM. SKGS mapping and SWMI-guided exploration form a closed loop to improve target-region reconstruction, while fine localization refines candidate camera poses for reliable goal-image matching and instance verification. The complete TAGSNav system achieves 71.60% SR and 26.99% SPL.

Consequently, these methods commonly rely on current RGB observations acquired from viewpoints constrained by robot embodiment and environmental traversability. Under these constraints, large viewpoint differences and visually similar distractors may prevent reliable instance correspondence, leading to false or missed matches [2], [4].

To address these limitations, recent studies have introduced 3D Gaussian Splatting (3DGS) [7]–[9] into IIN. By representing 3D geometry and fine-grained appearance with efficient novel-view rendering, 3DGS provides a shared scene representation for map construction and instance verification, offering an effective basis for instance-level localization. Nevertheless, IIN requires online incremental mapping from sparse observations under tight optimization budgets, whereas global high-fidelity 3DGS reconstruction relies on sufficient observations and extensive iterative optimization [7], [10]. Existing methods therefore struggle to achieve efficient online updates while preserving the fine-grained appearance details required for instance matching [8], [9]. In IIN, however, exhaustive reconstruction is neither affordable nor necessary—under a fixed optimization budget, only the appearance of target instances warrants faithful modeling, while the rest of the scene merely serves as geometric context. This raises the central question: *how can a target-aware Gaussian map be constructed online from sparse observations to support reliable instance matching from arbitrary viewpoints for navigation?*

To answer this, we propose TAGSNav, an online 3DGS navigation framework that integrates target-guided Gaussian reconstruction with instance localization. To improve target-

region reconstruction under sparse observations and online constraints, we develop Incremental Semantic Keyframe-Guided Gaussian Splatting (SKGS) mapping and an exploration strategy guided by Gaussian-Uncertainty-Driven Semantically Weighted Mutual Information (SWMI). SKGS mapping uses historical target-category keyframes and mask-weighted losses to incrementally refine target-relevant regions under a fixed computational budget. SWMI-guided exploration selects informative viewpoints based on the uncertainty of target-category Gaussians, better utilizing sparse observations. The two strategies form a closed loop (Fig. 1): SKGS updates target-category Gaussians, whose uncertainty scores are then recomputed from rendering residuals for SWMI; SWMI-selected viewpoints provide new observations for further SKGS refinement. To handle large viewpoint differences and similar distractors, we introduce coarse-to-fine instance localization with conditional PnP-based Viewpoint Alignment (PVA) for feedback correction. Compensatory re-exploration verifies uncertain candidates using additional observations, reducing false and missed matches.

The main contributions are summarized as follows:

- We propose TAGSNav, an online IIN framework that unifies target-aware online Gaussian mapping and instance verification within a shared scene representation.
- We develop SKGS mapping and SWMI-guided exploration to improve target-region reconstruction under sparse observations and limited computation.
- We develop coarse-to-fine instance localization with conditional PVA and compensatory re-exploration for reliable matching under large viewpoint differences and visually similar distractors.
- We validate TAGSNav through HM3D-SEM simulations and real-robot experiments, achieving state-of-the-art SR and SPL in simulation.

II. RELATED WORK

A. Instance ImageGoal Navigation

Instance ImageGoal Navigation (IIN) [2] generalizes ObjectNav [11]–[13], which seeks any instance of a specified category, and ImageNav [14], which navigates to a goal-image location. IIN instead requires reaching the instance shown in the goal image, thereby addressing ambiguity among same-category instances and better reflecting object-search needs. Existing methods follow two paradigms. End-to-end methods [4], [15] directly predict actions using learned policies without explicit mapping, limiting interpretability, controllability, and debugging. Modular methods [5], [6], [16] instead separate the task into mapping, exploration, localization, and navigation modules. Such a modular structure facilitates failure diagnosis and component-level improvement. Our approach follows this paradigm.

B. Scene Representations for Visual Navigation

In modular methods, map representation affects cross-view matching capability. Graph-based representations, including topological maps [17] and scene graphs [16], encode environments as nodes and edges to capture connectivity and

high-level semantics. However, these graph abstractions do not directly support photorealistic novel-view rendering for instance-level matching. Metric maps preserve spatial coordinates and physical structure. 2D occupancy and semantic maps [5], [13] lack 3D geometry and appearance, and thus are used mainly for spatial planning; target matching is consequently left to the current RGB observation [6], precluding arbitrary-viewpoint verification. TSDF maps [18] and octree occupancy maps [19] support online updates and encode 3D geometry, but their geometric representations do not directly support photorealistic novel-view rendering for appearance-based matching. NeRF-based representations [20] enable differentiable novel-view synthesis, but NeRF incurs substantial per-scene training and volumetric rendering costs. Generalizable neural radiance maps [21] enable incremental mapping without per-scene optimization, yet each rendered view still requires dense neural-field queries along camera rays, whose cost scales with image resolution.

C. 3DGS-Based Mapping for Navigation

3D Gaussian Splatting (3DGS) [7] has been adopted in image-guided navigation due to its real-time photorealistic novel-view rendering [8], [9], [22], [23]. Both feed-forward and optimization-based mapping have been explored for 3DGS-based image-guided navigation. Feed-forward methods [9] support online incremental mapping but lack iterative per-scene map refinement and IIN-oriented target-region refinement. Optimization-based methods iteratively refine scene maps to improve novel-view rendering quality, but existing optimization-based 3DGS navigation methods [8], [24] rely on preconstructed maps, requiring scene reconstruction before navigation. Although optimization-based online mapping supports per-scene refinement, its computational cost and dependence on sufficient observations hinder its use in online IIN. Coordinating optimization-budget allocation and viewpoint selection to improve target-relevant regions remains a challenge in online IIN. We address this by coupling target-aware map optimization with informative viewpoint selection to improve target-region reconstruction.

III. METHODOLOGY

A. Problem Formulation

We formulate IIN as a target-aware, budgeted, factorized decision process. Given a goal image I_g , the robot must reach the corresponding instance in an unknown environment using sequential RGB-D observations $o_t = (I_t, D_t, P_t)$, where I_t and D_t denote the RGB image and depth image at time t , respectively, and P_t denotes the estimated camera pose. The system maintains an online 3DGS map, a semantic occupancy map, candidate instances, and Gaussian uncertainty estimates. At time t , the system state is

$$s_t = (\mathcal{G}_t, \mathcal{M}_t, \mathcal{R}_t, \mathcal{U}_t, \mathcal{B}_t), \quad (1)$$

where $\mathcal{G}_t$ is the online 3DGS map, $\mathcal{M}_t$ is the semantic occupancy map, $\mathcal{R}_t$ is the candidate-instance registry, $\mathcal{U}_t$ stores

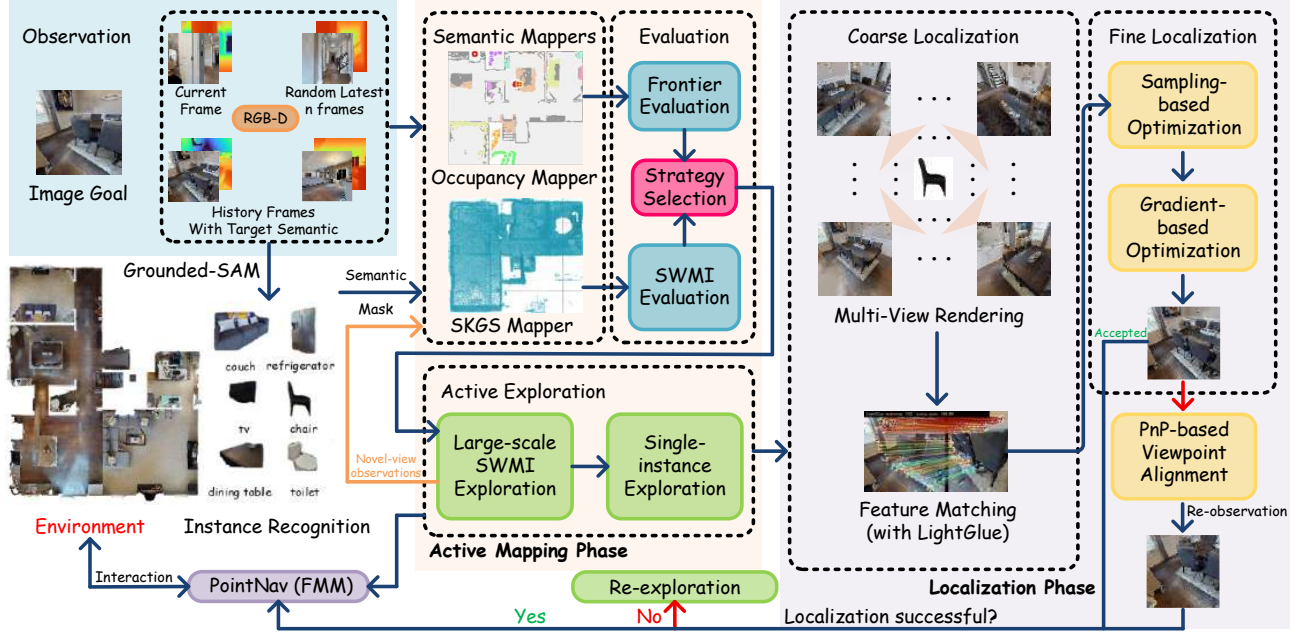


Fig. 2: Our framework comprises four components: mapping, exploration, localization, and re-exploration. Mapping and exploration form a closed loop. SKGS refines the online 3DGS map using target-category keyframes, while SWMI selects informative viewpoints based on Gaussian uncertainty. Coarse and fine localization operate jointly: the former finds correspondences via multi-view rendering and feature matching, whereas the latter refines the capture pose and conditionally invokes PnP-based viewpoint alignment for verification. When explored-area growth stalls and the target remains unlocalized, re-exploration revisits rejected candidates for supplementary observations.

per-Gaussian uncertainty estimates, and $\mathcal{B}_t$ buffers temporarily rejected candidates. Solving this process online is challenging due to sparse observations and a fixed optimization budget. TAGSNav therefore couples SKGS mapping with SWMI-guided exploration in a closed loop to improve the scene representation for target-instance localization.

This section presents the online mapping–exploration loop, coarse-to-fine localization with viewpoint alignment, conditional re-exploration, and final goal execution. The framework is shown in Fig. 2.

B. Online Scene Representation

1) *2D Semantic Occupancy Mapping:* At each time step, the RGB-D observation (I_t, D_t) is projected into world coordinates using pose P_t to update the semantic occupancy grid $\mathcal{M}_t$, which encodes traversability and observed target semantics. The grid supports FMM planning and provides reachable goals for exploration, localization, and navigation.

2) *Semantic Keyframe-Guided 3DGS Mapping:* TAGSNav incrementally updates the online 3DGS map using SKGS. At time t , the map is represented as

$$\mathcal{G}_t = \{(\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j, \mathbf{c}_j, o_j, s_j)\}_{j=1}^{N_t}, \quad (2)$$

where each Gaussian has a center, covariance, color, and opacity [7], augmented with target-category confidence s_j . Frames with target-category detections are selected as RGB-D keyframes $F_i = (I_i, D_i, P_i)$. Differentiable rendering of $\mathcal{G}_t$ from P_i predicts the RGB image $\hat{I}_i$ and depth map $\hat{D}_i$. The choice of optimization frames directly affects reconstruction quality for target matching.

Conventional online 3DGS methods optimize the map using the current frame or frames sampled from a sliding

window [10]. However, an IIN target may appear in only a few early views and never be observed again. Once these frames leave the window, their corresponding Gaussians receive no further optimization. TAGSNav uses Grounded-SAM [25] to extract the target category and goal-instance mask, stores masked RGB-D keyframes, and selects m viewpoint-diverse keyframes to form the historical set $\mathcal{H}_t^{\text{obj}}$.

To keep historical target observations involved in optimization, we divide the fixed budget of K optimization steps per update among three frame sources:

$$K = K_{\text{cur}} + K_{\text{win}} + K_{\text{hist}}, \quad (3)$$

where the terms on the right denote the steps allocated to the current frame, recent frames in $\mathcal{W}_t$, and historical target-category keyframes in $\mathcal{H}_t^{\text{obj}}$, respectively. This allocation enables continued refinement of past target observations without increasing the total number of optimization steps.

For keyframe F_i , let $S_i^{\text{obj}}(u) \in \{0, 1\}$ denote its target-category mask at pixel u . The pixel weight is

$$w_i(u) = 1 + \mathbb{I}(F_i \in \mathcal{H}_t^{\text{obj}}) \lambda_{\text{mask}} S_i^{\text{obj}}(u), \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function and λ_{mask} weights target regions. At each optimization step, we select a frame F_i according to the budget allocation above and optimize the Gaussian parameters by minimizing

$$\ell_i = \lambda_p \mathcal{L}_{\text{rgb}}^i(w_i) + \lambda_g \mathcal{L}_{\text{depth}}^i(w_i) + \lambda_{\text{iso}} \mathcal{L}_{\text{iso}}. \quad (5)$$

Here, $\mathcal{L}_{\text{rgb}}^i$ and $\mathcal{L}_{\text{depth}}^i$ are RGB and depth reconstruction losses weighted by w_i , $\mathcal{L}_{\text{iso}}$ regularizes Gaussian scales, and $\lambda_p, \lambda_g, \lambda_{\text{iso}}$ are their respective weights.

Within the fixed budget, SKGS preserves historical cross-view constraints to improve target-region reconstruction for

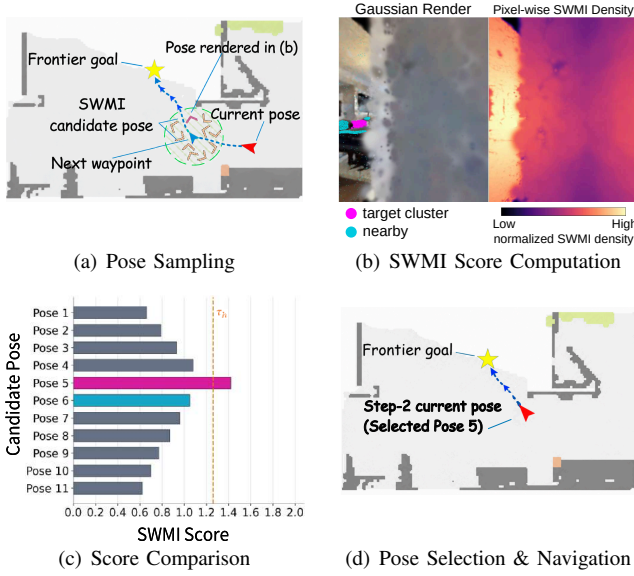


Fig. 3: One SWMI-guided exploration step. (a) N_{pose} collision-free poses are sampled around the next FMM waypoint σ_t^W , which is also included in $\mathcal{C}_t$. (b) Each candidate is scored using its 3DGS rendering and pixel-wise SWMI contributions; the highlighted pose in (a) is shown. (c) Relative SWMI scores compare candidate informativeness, highlighting the highest-scoring pose (pink) and the next waypoint (blue). (d) The selected pose becomes the next observation goal, and FMM plans a path to it.

localization, while rendering residuals determine uncertainty scores for subsequent SWMI exploration.

C. Mutual Information-Based Active Exploration Strategy

Since optimization cannot replace additional viewpoints, SWMI combines Gaussian uncertainty scores with semantic relevance to select viewpoints that observe high-uncertainty target-category regions and their neighborhoods.

1) *SWMI-Guided Large-Scale Exploration:* During online mapping, geometric frontier exploration efficiently explores unmapped space but does not actively revisit observed regions with incomplete reconstructions. In IIN, target-category regions and their neighborhoods may thus remain poorly reconstructed despite being visible in the map, impairing feature matching and localization. SWMI addresses this by directing future observations toward visible Gaussians with high reconstruction uncertainty and target relevance.

To evaluate the expected information gain from revisiting an observed region, we adopt the GauSS-MI score [26] for a candidate pose σ :

$$I_{\text{GM}}(\sigma) = \sum_{j=1}^{n_z} P(z_j | M_j) \sum_{i \in \mathcal{N}_{\sigma}^j} -T_{\sigma}^{i,j} \log P(r_i). \quad (6)$$

Here, j indexes the n_z pixel rays, $\mathcal{N}_{\sigma}^j$ is the set of Gaussians along ray j , $T_{\sigma}^{i,j}$ is the rendering weight of Gaussian i , and $P(r_i)$ is its reliability probability, so $-\log P(r_i)$ is the uncertainty score stored in $\mathcal{U}_t$; $P(z_j | M_j)$ is the sensor likelihood given the rendered luminance M_j . The score thus reflects only geometric reconstruction quality.

However, standard GauSS-MI does not prioritize target semantics, so high-uncertainty background Gaussians can dominate its score and divert observations from regions needed for matching and localization. We therefore assign three semantic and spatial weights. Gaussians with target-category confidence $s_i \geq \tau_s$ are clustered into candidate instances. Other Gaussians within a ground-plane distance ρ of the nearest instance center form the neighborhood; the rest are background. Here, τ_s and ρ are the confidence threshold and neighborhood radius. Their weights are $\lambda_{\text{obj}} > \lambda_{\text{near}} > \lambda_{\text{bg}} = 1$. Incorporating these weights as ω_i yields the SWMI score:

$$I_{\text{SWMI}}(\sigma) = \sum_{j=1}^{n_z} P(z_j | M_j) \sum_{i \in \mathcal{N}_{\sigma}^j} -\omega_i T_{\sigma}^{i,j} \log P(r_i). \quad (7)$$

This weighting prioritizes target regions and their context while retaining background contributions. The complete SWMI-guided pose-selection process is illustrated in Fig. 3.

At each decision step, the nearest reachable frontier f_t^{near} is selected based on path distance. FMM then generates the next waypoint $\sigma_t^W = (\mathbf{p}_t^W, \theta_t^W)$ along the frontier path, where $\mathbf{p}_t^W$ and θ_t^W denote its position and heading. To improve reconstruction while limiting detours from the path to the selected frontier, we sample N_{pose} collision-free candidate poses $\{\sigma_t^{\ell}\}_{\ell=1}^{N_{\text{pose}}}$ whose positions lie within a radius r of $\mathbf{p}_t^W$. Together with the FMM waypoint σ_t^W , these poses form

$$\mathcal{C}_t = \{\sigma_t^W\} \cup \{\sigma_t^{\ell}\}_{\ell=1}^{N_{\text{pose}}}. \quad (8)$$

Here, ℓ indexes the locally sampled poses; therefore, $\mathcal{C}_t$ contains $N_{\text{pose}} + 1$ candidate poses. Their positions are constrained to traversable space, while their orientations are sampled and evaluated using the SWMI score in Eq. (7).

Among the candidates in $\mathcal{C}_t$, the best pose and the SWMI scores of this pose and the FMM waypoint are defined as

$$\begin{aligned} \sigma_t^* &= \arg \max_{\sigma \in \mathcal{C}_t} I_{\text{SWMI}}(\sigma), \\ S_t^* &= I_{\text{SWMI}}(\sigma_t^*), \quad S_t^W = I_{\text{SWMI}}(\sigma_t^W). \end{aligned} \quad (9)$$

Given Eq. (9), the next exploration goal is

$$\sigma_t^{\text{goal}} = \begin{cases} \sigma_t^*, & S_t^* \geq \tau_h \wedge S_t^W < \tau_l, \\ \sigma_t^W, & \text{otherwise,} \end{cases} \quad \tau_h > \tau_l. \quad (10)$$

The thresholds $\tau_h > \tau_l$ ensure that a detour is taken only when the best candidate is sufficiently informative and the FMM waypoint has low informativeness. This rule maintains efficient progress toward f_t^{near} while directing observations toward high-uncertainty target-category Gaussians and their neighborhoods to improve subsequent localization and matching. SWMI selects the next observation pose rather than optimizing the full trajectory; FMM plans the path to the selected goal, and the localization module verifies the target instance. This division keeps viewpoint selection local and computationally tractable.

At the selected viewpoint, the robot acquires an RGB-D observation to update the SKGS map, retaining selected

target-category keyframes for subsequent historical optimization. After the map update, rendering errors refresh the Gaussian uncertainty scores used in the next SWMI decision, completing the mapping–exploration loop.

2) *Instance Confirmation and Single-Instance Exploration*: During large-scale frontier exploration, when target-category detections persist for n_c consecutive frames, N_c fixed-radius poses $\{\sigma_l\}_{l=1}^{N_c}$ are sampled around the center of each cluster $\mathcal{G}_k^{\text{obj}}$, and the rendered views are matched with the goal image using the semantically weighted matching score $S_k(\sigma_l)$, as defined later in Eq. (11). Let

$$s_k = \max_l S_k(\sigma_l), \quad k^* = \arg \max_k s_k.$$

If $s_{k^*} \geq \tau_m$, the highest-scoring cluster is accepted as a candidate; otherwise, frontier exploration resumes. A lower threshold τ_m admits preliminary candidates, which are subsequently verified using stricter thresholds τ_c and τ_f in coarse and fine localization, respectively. For the accepted candidate, the robot samples poses within $\alpha_r r_{k^*}^{\text{obj}}$ of its center, where $\alpha_r > 1$ is an expansion factor and $r_{k^*}^{\text{obj}}$ is the radius of $\mathcal{G}_{k^*}^{\text{obj}}$. It selects the pose with the highest SWMI score and uses FMM to approach it, acquiring supplementary RGB-D observations en route and at the selected pose. The local 3DGS map is updated before subsequent localization.

D. Coarse-to-Fine Localization and Viewpoint Alignment

Using the SKGS map refined by SWMI-guided observations, localization resolves instance ambiguity and viewpoint differences through coarse matching, pose refinement, and optional viewpoint alignment (Fig. 4). Preliminary instance confirmation and all localization stages evaluate rendered views using the same semantically weighted matching score. Specifically, the rendering of candidate instance k from pose σ is matched against the goal image using SuperPoint–LightGlue [27], [28]:

$$S_k(\sigma) = \sum_{p \in \mathcal{P}_k(\sigma)} \alpha_p \gamma_p, \quad \alpha_p = \begin{cases} \eta_{\text{sem}}, & \mathbf{x}_p^{\text{goal}} \in \mathcal{M}^{\text{goal}}, \\ \eta_{\text{bg}}, & \text{otherwise,} \end{cases} \quad (11)$$

where $\mathcal{P}_k(\sigma)$ is the set of valid matches, γ_p is the confidence of match p , $\mathbf{x}_p^{\text{goal}}$ is its coordinate in the goal image, and $\mathcal{M}^{\text{goal}}$ is the goal-instance mask. With $\eta_{\text{sem}} > \eta_{\text{bg}} > 0$, in-mask matches are emphasized while out-of-mask matches provide auxiliary context.

1) *Coarse Localization*: At the SWMI-selected observation pose, target-category masks from the current RGB image are back-projected onto the 3DGS map using depth consistency, updating the candidate instance clusters $\{\mathcal{G}_k^{\text{obj}}\}$. For each cluster, virtual camera poses sampled at multiple scales and viewing angles form $\mathcal{V}_k$. Its best coarse pose is

$$\sigma_k^{\text{coarse}} = \arg \max_{\sigma \in \mathcal{V}_k} S_k(\sigma).$$

The cluster and its coarse pose are retained if $S_k(\sigma_k^{\text{coarse}}) \geq \tau_c$, where τ_c is the coarse-localization threshold; otherwise, the cluster is rejected.

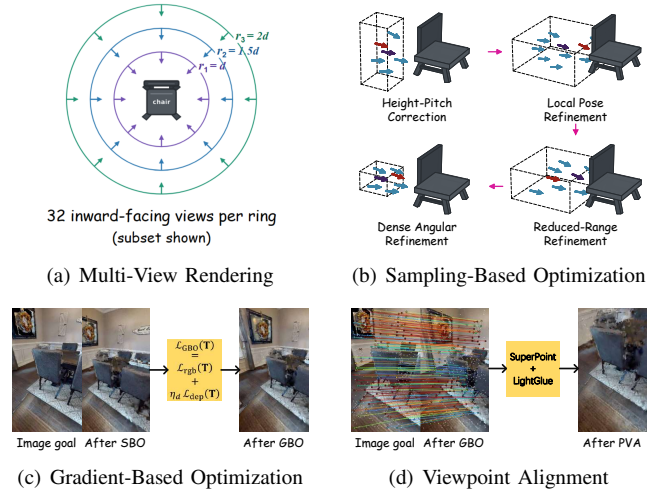


Fig. 4: Coarse-to-fine localization with conditional PVA. The first two stages identify a suitable viewpoint and progressively refine its camera pose. In the final two stages, differentiable rendering further optimizes the pose, while PVA is invoked only when additional viewpoint correction is needed.

2) *Fine Localization*: Coarse matching may confuse adjacent same-category instances and produce a biased matching viewpoint. We therefore refine each retained pose using sampling-based optimization (SBO) followed by gradient-based optimization (GBO). SBO performs a coarse-to-fine search over position, yaw, and pitch through four stages of rendering and matching, which successively correct height and pitch, refine the local pose, repeat the search over a reduced range, and densely optimize the final viewing angles. At each stage, the best pose is retained only if it increases the matching score in Eq. (11). The resulting SBO pose initializes GBO. GBO optimizes the pose $\mathbf{P}$ with Adam using differentiable rendering:

$$\mathcal{L}_{\text{GBO}} = \mathcal{L}_{\text{rgb}} + \eta_d \mathcal{L}_{\text{dep}}. \quad (12)$$

In Eq. (12), $\mathcal{L}_{\text{rgb}}$ and $\mathcal{L}_{\text{dep}}$ are the masked RGB and depth residuals, respectively, and η_d weights the depth term. The depth term is omitted when goal depth is unavailable. Following LoGS [29], M_{grad} , M_{fea} , and M_{occ} denote masks for pixels with strong gradients, SuperPoint keypoint neighborhoods, and reliable Gaussian coverage, respectively. Their combination $M = (M_{\text{grad}} \cup M_{\text{fea}}) \cap M_{\text{occ}}$ retains informative pixels with reliable coverage. The refined pose and matching statistics are used to accept, align, or reject the candidate. The final navigation pose is derived from the selected instance cluster rather than from the optimized matching pose.

3) *PnP-Based Viewpoint Alignment*: A high matching score alone may not confirm the target because background correspondences or a visually similar instance can dominate the score. After GBO, the system re-renders at the optimized pose $\mathbf{P}_{\text{GBO}}$, recomputes S_{GBO} using Eq. (11), and counts the in-mask matches N_{sem} . With $\tau_f > \tau_c$, the result is accepted if $S_{\text{GBO}} \geq \tau_f$ and $N_{\text{sem}} \geq \tau_{\text{sem}}$, and rejected if $S_{\text{GBO}} < \tau_f$. Here, τ_f and τ_{sem} denote the score and in-mask match thresholds, respectively. When $S_{\text{GBO}} \geq \tau_f$ but $N_{\text{sem}} < \tau_{\text{sem}}$, we use the camera-position change $\Delta_p = \|\mathbf{p}_{\text{GBO}} - \mathbf{p}_{\text{coarse}}\|_2$

as an empirical criterion for additional verification, where $\mathbf{p}_{\text{GBO}}$ and $\mathbf{p}_{\text{coarse}}$ are the world-frame camera centers of the GBO and coarse poses, respectively. PVA is triggered if $\Delta_p > L$, where L is the displacement threshold; otherwise, the localization result is accepted. This additional condition avoids triggering PVA solely because of limited in-mask matches, which may arise from weak target texture. PVA back-projects cached rendered-view keypoints using rendered depth to form 2D–3D correspondences for RANSAC-PnP [30], [31], preferring in-mask correspondences and using full-image ones only when foreground matches are insufficient. The robot then moves to the corrected observation pose, updates the local map, repeats SBO–GBO refinement, and begins final navigation after localization is confirmed.

A rejected candidate at any stage is placed in $\mathcal{B}_t$ for κ frames, where κ is the blacklist duration. After expiry, it returns to $\mathcal{R}_t$, as continued mapping may have improved its Gaussian representation.

E. Re-exploration

When map coverage is nearly saturated but the target remains unlocalized, frontier exploration may drift toward peripheral areas instead of revisiting previously detected candidates. The re-exploration module revisits these candidates for supplementary verification.

1) *Triggering*: Re-exploration starts when explored-area growth over a window of W_g frames falls below T_g while the target remains unlocalized. Here, W_g is the window length and T_g is the minimum exploration-growth threshold. If growth recovers, the system resumes frontier exploration.

2) *Candidate Selection*: Eligible candidates are target-category instances detected during initial exploration with sufficient mask area, an FMM-reachable approach point, and no active blacklist entry. They are visited in increasing path-cost order, with larger masks breaking ties.

3) *Candidate Confirmation*: Once visible, a revisited candidate is reconfirmed. If confirmed, the robot proceeds to single-instance exploration, localization, and final navigation (Secs. III-C.2, III-D, and III-F); otherwise, the candidate is temporarily blacklisted and the next is tested. Re-exploration ends when the target is localized or all candidates have been tested once; in the latter case, the task is unsuccessful.

F. Navigation

Once target localization is confirmed, target-category Gaussians within the estimated camera frustum and projecting inside the goal-instance mask are clustered using DBSCAN [32]. After filtering by volume, the cluster closest to the camera center is selected as the target, and its centroid is projected onto the 2D semantic occupancy grid. FMM plans to this cell if free and reachable; otherwise, it plans to the nearest reachable free cell.

IV. EXPERIMENTS

This section evaluates TAGSNav in simulation using the Habitat-Matterport 3D Semantics (HM3D-SEM) dataset [33] and on a physical robot. We present the experimental setup

TABLE I: Key parameter settings used in our experiments

Module	Parameter	Value
SKGS	Optimization budget K	12
	Steps per source (cur./win./hist.)	3/6/3
	Mask weight λ_{mask}	1.2
SWMI	$\lambda_{\text{obj}}/\lambda_{\text{near}}/\lambda_{\text{bg}}$	2/1.5/1
Loc.	Rings $\times$ views per ring	3×32
	Ring radii (cluster diam.)	$1/1.5/2 \times$
	$\tau_m/\tau_c/\tau_f$	20/40/70

cur./win./hist. denote the current-frame, recent-window, and historical-keyframe sources, respectively; cluster diam. denotes multiples of the target-cluster diameter. Module descriptions are given in Sec. III.

and metrics, comparisons with state-of-the-art (SOTA) methods, ablation studies, and real-world results.

A. Experiments on the HM3D-SEM Dataset

1) *Experimental Setup and Evaluation Metrics*: We conduct experiments in the Habitat simulator [34] on the HM3D-SEM dataset [33], following the IIN Free-view protocol [5], where goal images are captured from arbitrary camera poses. We use 7,056 training episodes for development and hyperparameter tuning, and 1,000 validation episodes for evaluation. Target instances belong to six categories: {chair, couch, plant, bed, toilet, tv-monitor}. All experiments use the hyperparameters listed in Table I.

Metrics: We report SR and SPL [12], measuring success frequency and success-weighted path efficiency, respectively; higher is better. An episode succeeds when the agent stops at a Euclidean distance below 1.0 m from the target instance’s geometric center. Fig. 5 shows a complete simulated episode and cross-view localization examples.

2) *Comparative Analysis with Existing Methods*: All methods operate online without prebuilt maps. We use published HM3D-SEM results and detail the two strongest baselines below:

IEVE. IEVE [6] is an IIN method based on deep reinforcement learning (DRL). It switches among exploration, verification, and exploitation using feature matches and candidate distances, approaching targets before navigation.

UniGoal. UniGoal [16] is a zero-shot modular framework that represents observations and goals as graphs and matches the scene graph with the goal graph at each time step. It uses LLaMA-2-7B reasoning for ObjectNav, IIN, and TextNav.

Table II compares TAGSNav with IEVE, UniGoal, and two earlier baselines, OVRL-v2-IIN [5], [35] and Mod-IIN [5], on HM3D-SEM. TAGSNav achieves SOTA results in both SR and SPL.

TAGSNav outperforms all baselines, including IEVE and UniGoal. Although IEVE uses sequential approach-to-confirm verification and UniGoal employs scene–goal graph reasoning, their final verification still relies on local-feature matching between the current RGB observation and goal image. Consequently, they must approach each candidate for a matchable view. In contrast, TAGSNav leverages an optimized high-quality 3DGS map to render and match candidates from resampled virtual viewpoints, enabling distant confirmation without visiting each viewpoint. This remote

In-situ 3DGS renderings during the episode process (re-exploration was not triggered)



Successful localization with a large viewpoint gap between the RGB observation and the goal image



Fig. 5: Qualitative HM3D-SEM results. Top: the TAGSNav pipeline from exploration and online 3DGS mapping to target localization and navigation. Bottom: successful localization despite large viewpoint gaps between the observation and goal image.

TABLE II: Comparison of IIN methods on HM3D-SEM

Paradigm	Method	SR (%) $\uparrow$	SPL (%) $\uparrow$
End-to-end	OVRL-v2-IIN	24.80	11.80
Modular	Mod-IIN	56.10	23.30
	UniGoal	60.20	23.70
	IEVE	70.20	25.20
	TAGSNav (Ours)	71.60	26.99

Higher is better ($\uparrow$). Best and second-best results are shown in **bold** and underlined, respectively.

TABLE III: Module ablation on HM3D-SEM (100 episodes)

Variant	SKGS	SWMI	FL+PVA	RE	SR (%) $\uparrow$	SPL (%) $\uparrow$
w/o SKGS and SWMI			$\checkmark$	$\checkmark$	57	21.90
w/o SKGS		$\checkmark$	$\checkmark$	$\checkmark$	62	23.52
w/o SWMI	$\checkmark$		$\checkmark$	$\checkmark$	64	24.12
w/o FL+PVA	$\checkmark$	$\checkmark$		$\checkmark$	64	26.60
w/o RE	$\checkmark$	$\checkmark$	$\checkmark$		66	27.13
TAGSNav	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	71	28.22

$\checkmark$: module used. FL+PVA: fine localization with conditional PnP-based viewpoint alignment; RE: re-exploration. $\uparrow$: larger is better; **bold**: best.

verification can reduce unnecessary travel and improve navigation efficiency. Together, target-focused reconstruction and arbitrary-view synthesis provide reliable cross-view localization and superior navigation performance.

3) *Ablation Study:* To reduce computational cost, we evaluate all variants on the same set of 100 episodes sampled from the HM3D-SEM validation set. All variants use 12 optimization steps per mapping update. Table III evaluates the four components and the combined SKGS–SWMI effect.

w/o SKGS and SWMI. Mapping uses only a recent-frame sliding window for online map optimization, while exploration follows the nearest-frontier strategy.

w/o SKGS. The 3DGS map is optimized using only a recent-frame sliding window, while SWMI-guided exploration is retained throughout the navigation process.

w/o SWMI. Exploration reverts to the nearest-frontier strategy, while SKGS mapping is retained.

w/o Fine Localization and PVA. The coarse pose is used directly to extract the target cluster and navigation goal, without subsequent pose refinement or viewpoint verification.

w/o Re-exploration. The agent continues SWMI-guided frontier exploration without targeted candidate revisits.

Removing either SKGS or SWMI reduces SR and SPL, validating historical target-region optimization and informative viewpoint selection, while removing both produces the lowest performance. Relative to this baseline, their combined gains exceed the summed individual gains by 2.0 percentage points for SR and 2.48 for SPL, indicating synergy. The SR reductions after removing FL+PVA and re-exploration

support their roles in target confirmation and failure recovery; both removals have smaller effects on SPL than removing SKGS or SWMI. These results validate the framework.

B. Real-World Experiments

We deploy TAGSNav without additional training or fine-tuning on a modified mobile chassis in an indoor laboratory. Four missions vary in target distance and occlusion: Scene 1 contains an unoccluded long-range chair search, whereas Scene 2 contains an unoccluded short-range chair search, an occluded short-range plant search, and an occluded long-range chair search (Fig. 6). Similar chair distractors test instance discrimination; the elevated plant and supporting table test mapping and safe planning. TAGSNav stops within 1 m in all chair missions. In the plant mission, it identifies the target and reaches the nearest safe cell, approximately 1.2 m away because of obstacle inflation around the table. This demonstrates recognition and safe approach but does not meet the 1-m simulation criterion. These results provide preliminary evidence of direct deployment and Sim2Real transfer. Videos are available on the project webpage.

V. CONCLUSIONS

In this paper, we presented TAGSNav, a target-aware IIN framework that unifies online 3DGS mapping, active exploration, and instance localization. SKGS refines target-category regions using historical observations, while SWMI guides future observations toward informative viewpoints, forming a closed mapping–exploration loop. Coarse-to-fine localization, conditional PVA, and compensatory

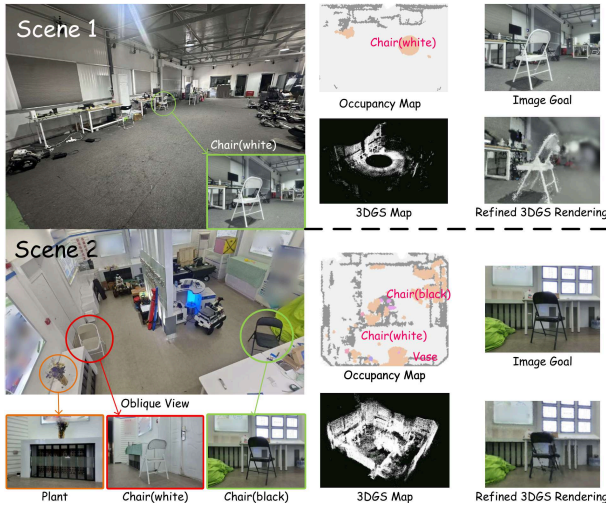


Fig. 6: Qualitative real-world results for chair and plant targets.

re-exploration further reduce false and missed matches. TAGSNav achieved higher SR and SPL than existing methods on HM3D-SEM. Real-robot experiments without additional training or fine-tuning demonstrated the feasibility of direct deployment and provided preliminary evidence of Sim2Real transfer. Future work will reduce computational latency in Gaussian map optimization, instance verification, and localization to better support real-time navigation.

REFERENCES

- [1] Y. Tang *et al.*, “Openin: Open-vocabulary instance-oriented navigation in dynamic domestic environments,” *IEEE Robotics and Automation Letters*, vol. 10, no. 9, pp. 9256–9263, 2025.
- [2] J. Krantz *et al.*, “Instance-specific image goal navigation: Training embodied agents to find object instances,” *arXiv preprint arXiv:2211.15876*, 2022.
- [3] Y. Tang *et al.*, “OpenObject-NAV: Open-vocabulary object-oriented navigation based on dynamic carrier-relationship scene graph,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 20 182–20 188.
- [4] G. Bono *et al.*, “End-to-end (Instance)-Image goal navigation through correspondence as an emergent phenomenon,” in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024. [Online]. Available: <https://openreview.net/forum?id=cphhnHjCvC>
- [5] J. Krantz *et al.*, “Navigating to objects specified by images,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Oct. 2023, pp. 10 916–10 925.
- [6] X. Lei *et al.*, “Instance-aware exploration-verification-exploitation for instance ImageGoal navigation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 16 329–16 339.
- [7] B. Kerbl *et al.*, “3D Gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, pp. 1–14, Jul. 2023.
- [8] X. Lei *et al.*, “GaussNav: Gaussian splatting for visual navigation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 47, no. 5, pp. 4108–4121, May 2025.
- [9] W. Guo *et al.*, “IGL-Nav: Incremental 3D Gaussian localization for image-goal navigation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025, pp. 6808–6817.
- [10] H. Matsuki *et al.*, “Gaussian splatting SLAM,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 18 039–18 048.
- [11] J. Sun *et al.*, “A survey of object goal navigation,” *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 2292–2308, 2025.
- [12] P. Anderson *et al.*, “On evaluation of embodied navigation agents,” *arXiv preprint arXiv:1807.06757*, 2018. [Online]. Available: <https://arxiv.org/abs/1807.06757>
- [13] D. S. Chaplot *et al.*, “Object goal navigation using goal-oriented semantic exploration,” in *Advances in Neural Information Processing Systems*, vol. 33. Curran Associates, Inc., 2020, pp. 4247–4258.
- [14] Y. Zhu *et al.*, “Target-driven visual navigation in indoor scenes using deep reinforcement learning,” in *2017 IEEE International Conference on Robotics and Automation (ICRA)*, 2017, pp. 3357–3364.
- [15] X. Sun *et al.*, “FGPrompt: Fine-grained goal prompting for image-goal navigation,” in *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [16] H. Yin *et al.*, “UniGoal: Towards universal zero-shot goal-oriented navigation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 19 057–19 066.
- [17] X. Cui *et al.*, “Frontier-enhanced topological memory with improved exploration awareness for embodied visual navigation,” in *Computer Vision – ECCV 2024*. Springer, 2024, pp. 296–313.
- [18] R. A. Newcombe *et al.*, “KinectFusion: Real-time dense surface mapping and tracking,” in *2011 10th IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2011, pp. 127–136.
- [19] A. Hornung *et al.*, “OctoMap: An efficient probabilistic 3D mapping framework based on octrees,” *Autonomous Robots*, vol. 34, no. 3, pp. 189–206, 2013.
- [20] B. Mildenhall *et al.*, “NeRF: Representing scenes as neural radiance fields for view synthesis,” in *Computer Vision – ECCV 2020*. Springer, 2020, pp. 405–421.
- [21] O. Kwon, J. Park, and S. Oh, “Renderable neural radiance map for visual navigation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2023, pp. 9099–9108.
- [22] D. Charatan *et al.*, “pixelSplat: 3D Gaussian splats from image pairs for scalable generalizable 3D reconstruction,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 19 457–19 467.
- [23] Y. Chen *et al.*, “MVSplat: Efficient 3D Gaussian splatting from sparse multi-view images,” in *Computer Vision – ECCV 2024*. Springer, 2024, pp. 370–386.
- [24] W. Meng *et al.*, “BEINGS: Bayesian embodied image-goal navigation with Gaussian splatting,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 5252–5258.
- [25] T. Ren *et al.*, “Grounded SAM: Assembling open-world models for diverse visual tasks,” *arXiv preprint arXiv:2401.14159*, 2024.
- [26] Y. Xie *et al.*, “GauSS-MI: Gaussian splatting Shannon mutual information for active 3D reconstruction,” in *Robotics: Science and Systems XXI*, 2025.
- [27] D. DeTone, T. Malisiewicz, and A. Rabinovich, “SuperPoint: Self-supervised interest point detection and description,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, Jun. 2018.
- [28] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, “LightGlue: Local feature matching at light speed,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Oct. 2023, pp. 17 627–17 638.
- [29] Y. Cheng *et al.*, “LoGS: Visual localization via Gaussian splatting with fewer training images,” in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2025, pp. 15 029–15 036.
- [30] M. A. Fischler and R. C. Bolles, “Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography,” *Communications of the ACM*, vol. 24, no. 6, pp. 381–395, 1981.
- [31] V. Lepetit, F. Moreno-Noguer, and P. Fua, “EPnP: An accurate $O(n)$ solution to the PnP problem,” *International Journal of Computer Vision*, vol. 81, no. 2, pp. 155–166, 2009.
- [32] M. Ester *et al.*, “A density-based algorithm for discovering clusters in large spatial databases with noise,” in *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*. AAAI Press, 1996, pp. 226–231.
- [33] K. Yadav *et al.*, “Habitat-Matterport 3D semantics dataset,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, Jun. 2023, pp. 4927–4936.
- [34] M. Savva *et al.*, “Habitat: A platform for embodied AI research,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, Oct. 2019, pp. 9338–9346.
- [35] K. Yadav *et al.*, “OVRL-V2: A simple state-of-art baseline for ImageNav and ObjectNav,” *arXiv preprint arXiv:2303.07798*, 2023.